Sugarcane bagasse cellulose-derived hydroxyethyl cellulose-based biocomposite films reinforced with curcumin extracts for skin wound care applications

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Highlights

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Sugarcane bagasse cellulose serves as a sustainable base for biocomposite films.

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Curcumin extract enhances antioxidant activity and improves wound healing ability.

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Crosslinked HEC/PVA/SA/gelatin films exhibit controlled swelling for wound care.

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Synthesized Films displayed antibacterial effects and cytocompatibility in wound healing.

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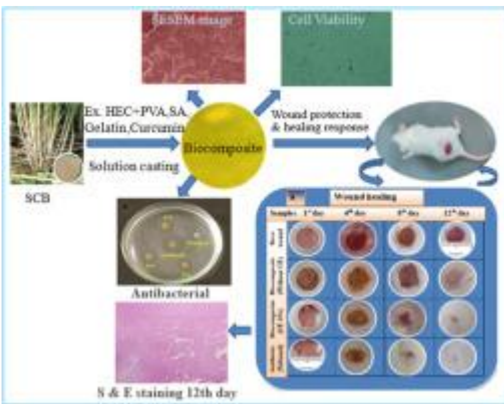
Developed films provide a biodegradable alternative to synthetic wound dressings.

Abstract

Biopolymer-based materials have attracted growing interest in wound healing due to their biocompatibility, biodegradability and regenerative potential. Sugarcane bagasse cellulose-derived hydroxyethyl cellulose-basedantibacterial biocomposite film wasdeveloped and evaluated through the crosslinking of hydroxyethyl cellulose (HEC), polyvinyl alcohol (PVA), succinic acid (SA), gelatin, and curcumin extract via solution casting. FTIR analysis confirmed the functional group interactions among the film components, while SEM images revealed the uniform dispersion of curcumin within the matrix. The films exhibited excellent mechanical properties, including tensile strength of 13.50 ± 0.10 MPa, swelling capacity of 1390% at pH 7.4, and water vapor transmission rate (WVTR) of 2330 g/m^2/day. Antibacterial tests showed inhibition zones of 16.15 mm for *S.*

aureus and 15.10 mm for *E. coli*, while antioxidant activity increased with curcumin content. Cytotoxicity assays on fibroblast cells showed high viability (95%) after 72 h. In vivo wound healing studies on mice revealed 95% wound closure within 12 days for curcumin-loaded films, compared to 70% (without CE loaded films) and 40% (Bare wound) groups. The films also promoted collagen deposition and reduced inflammation. These findings indicate that sugarcane bagasse-derived biocomposite films reinforced with curcumin hold strong potential as sustainable and effective wound dressing materials for biomedical applications.

Graphical abstract



1. [Download: Download high-res image (294KB)]
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Introduction

Wound care remains a critical area of research in biomedical science due to the challenges associated with chronic wounds, delayed healing and the risk of infections. Effective wound management materials are required to provide a protective barrier, maintain a moist environment and promote tissue regeneration while combating microbial infections [1]. Recent advances in wound-care technology have highlighted biopolymer-based materials as attractive candidates, mainly because they are naturally biocompatible, environmentally degradable and capable of carrying a wide range of therapeutic substances [2,3].

There has been a noticeable rise in the use of natural fiber–based biocomposites within biomedical research, especially in areas such as wound care, tissue regeneration and controlled drug delivery. Among these, cellulose-based bionanocomposites have received special attention because they offer an eco-friendly alternative to conventional synthetic materials. Their natural degradability and good interaction with living tissues make cellulose materials particularly appealing for medical applications [4].

Globally, there is growing interest in the development of eco-friendly and biodegradable biocomposite films for biomedical applications. Sugarcane bagasse, in particular, has emerged as a sustainable and abundant source for producing cellulose derivatives [5].

Sugarcane bagasse (SCB), the fibrous residue left after juice extraction in the sugar industry, represents a rich and renewable source of cellulose. Containing roughly 40–50% cellulose, SCB is widely recognized as a valuable raw material for producing cellulose derivatives, biocomposites and other bio-based materials. Its abundance, low cost and widespread availability in tropical and subtropical regions make it an attractive candidate for sustainable material development, particularly in biomedical fields. Moreover, the utilization of SCB adds value to agricultural waste and supports the global shift toward environmentally friendly and biodegradable materials [6].

As a fibrous byproduct of the sugar industry, sugarcane bagasse has emerged as an important and renewable source of cellulose for producing biocomposites. Its rich cellulose content, together with its widespread availability, makes it an appealing raw material for preparing cellulose derivatives used in biomedical materials and other applications [7]. In recent years, researchers have worked on modifying SCB-derived cellulose to improve its functional properties for biocomposite fabrication. These modifications are aimed at enhancing compatibility and performance, especially in biomedical applications [8].

Sugarcane bagasse, an abundant agricultural residue, has emerged as a promising raw material for extracting cellulose used in the synthesis of various biopolymers [5,9]. Cellulose derived from this biomass like sugarcane bagasse, can be converted into valuable derivatives such as hydroxyethyl cellulose (HEC). HEC is widely utilized in biomedical applications owing to its excellent film-forming capacity, strong water retention ability and favorable biocompatibility [10].

Polyvinyl alcohol (PVA) is widely utilized as a matrix in biocomposite films due to its outstanding ability to form uniform films. In addition, to enhancing the mechanical strength and flexibility of the films, PVA creates a smooth and continuous matrix that holds natural fibers or fillers in place [11].

Polyvinyl alcohol (PVA) possesses hydrophilic properties, which make it desirable for uses requiring water solubility. Although the biodegradability of synthetic PVA is limited to certain environmental conditions, its combination with natural polymers produces sustainable films that exhibit faster degradation than conventional plastic materials [12,13].

Polyvinyl alcohol (PVA) blends effectively with biopolymers, forming homogeneous films with enhanced physical, chemical and biodegradable properties. This synergy improves both the performance and sustainability of the resulting materials. PVA is frequently combined with hydroxyethyl cellulose (HEC) to increase the mechanical strength and stability of biocomposites [[14], [15], [16]], while succinic acid (SA) is employed as a cross-linking agent to further enhance structural integrity and biodegradability, making the films more suitable for sustainable biomedical applications. Succinic acid also contributes to improved thermal stability, mechanical strength and barrier properties and functions as an effective plasticizer in cellulose or chitosan-based films, enhancing flexibility and mechanical performance [17,18]. Additionally, the incorporation of gelatin, a natural protein polymer, increases the bioactivity of the composite by promoting cell adhesion, proliferation and tissue regeneration. Gelatin has attracted considerable attention in biocomposite development due to its excellent film-forming ability, biodegradability, edibility and inherent properties, including transparency, flexibility and effective oxygen barrier characteristics [[19], [20], [21], [22]].

Curcumin, a natural polyphenolic compound extracted from *Curcuma longa* (turmeric), has been widely recognized for its strong antioxidant, anti-inflammatory and antimicrobial properties. These bioactivities are particularly advantageous in wound healing, as they help reduce oxidative stress, control inflammation and prevent infections, thereby promoting faster tissue repair. However, the hydrophobic nature and limited stability of curcumin pose challenges for its direct application. Incorporating curcumin into biocomposite films offers a promising strategy, providing a stable and sustained delivery system that overcomes these limitations [[23], [24], [25], [26], [27]].

Biopolymer-based hydrogels have attracted considerable interest since the development of “superabsorbent” polymers in the 1950s. Over the past seventy years, significant research efforts from both academic and industrial sectors have explored their potential applications in various industries [28]. Hydrogels are highly regarded as exceptional absorbent materials due to their insolubility combined with the ability to absorb water in quantities thousands of times their dry weight. Their distinctive attributes, such as high water content, flexibility and biocompatibility, make them versatile for a wide range of applications, including biomedicine, biosensors, tissue engineering, drug delivery and environmental engineering [29].

Several studies have investigated sugarcane bagasse-based hydrogels with valuable findings, yet there are evident limitations. Pan et al. developed guanidine-grafted hydrogels with high swelling (∼2000%) and strong antimicrobial activity, but lacked therapeutic agents [30]. Mondal et al. synthesized highly adsorptive hydrogels (228 g/g) for pollutant removal but did not explore biomedical relevance [9]. Baiya et al. [31] presented green-synthesized CMC hydrogels with selective adsorption but lacked mechanical and biological evaluations. Islam et al.[23]developed biodegradable bagasse graft-copolymer hydrogels (swelling ~50 g/g) but omitted therapeutic functionality. Lastly, Ramphul et al. [32] reported sugarcane bagasse cellulose scaffolds with angiogenic potential but did not incorporate bioactive agents or fully evaluate mechanical, WVTR and biodegradation properties.

It is evident that the literature mentioned-above has certain limitations on its suitability as materials for good wound dressings. An ideal wound dressing material would have optimal mechanical strength, porosity, and biodegradability, as well as antibacterial, antioxidant, and wound-healing qualities, making it a promising sustainable material [32]. For this, the present research aimed to bridge the gaps in existing literature by integrating therapeutic functionality, biocompatibility and mechanical performance in curcumin-reinforced HEC/PVA/SA/Gelatin biocomposites. To the best of our knowledge, no studies have been conducted on the development of HEC, PVA, SA, Gelatin and CE biocomposites. The incorporation of curcumin extract into biocomposite films significantly enhances their total phenolic and flavonoid content, which correlates directly with increased antioxidant activity. For instance, Chitosan–PVA films loaded with curcumin demonstrated notable improvements in total phenolic and flavonoid content, enhancing radical scavenging and wound healing properties, thereby highlighting curcumin's role as a potent bioactive additive that elevates the functional performance of polymer-based wound dressings [33].

This research explores the fabrication of biocomposite films utilizing hydroxyethyl cellulose (HEC) extracted from sugarcane bagasse, combined with polyvinyl alcohol (PVA), succinic acid (SA) and gelatin, and reinforced with curcumin extract. The main objective of this research is to develop a multifunctional wound dressing that combines the mechanical strength of biopolymers with the therapeutic benefits of curcumin. The resulting biocomposites were systematically evaluated for their physicochemical properties, tensile strength, antibacterial activity and wound-healing efficacy through both in vitro and in vivo assessments. The findings provide valuable insights into the potential of these materials for advanced wound-care applications.

Materials

Sugarcane bagasse was obtained from the laboratory of Rajshahi Sugar Mills. The sugarcane was first crushed using a mini laboratory crusher and the resulting bagasse served as the primary raw material for cellulose extraction. Hydroxyethyl cellulose (HEC) was subsequently synthesized from the extracted cellulose. Polyvinyl alcohol (PVA), succinic acid (SA) and gelatin were procured from SRL, India. Turmeric (*Curcuma longa*) was procured from a local market. *S. aureus* and *E. coli* were collected

Characterization of extracted cellulose and biocomposite films

The extracted cellulose from sugarcane bagasse was characterized using Fourier transform infrared spectroscopy (FTIR) and X-ray diffraction (XRD) to confirm the removal of non-cellulosic components and evaluate its structural properties. FTIR spectra were recorded in the range of 4000–500 cm^{-1} using an FTIR spectrophotometer to identify the functional groups and chemical structure of the extracted cellulose. XRD analysis was performed using Cu Kα radiation (λ = 1.5406 Å) over a 2θ range of

Extracted cellulose characterization

The FTIR spectrum of the extracted cellulose (Fig. 5 a) confirmed the successful removal of lignin and hemicellulose from sugarcane bagasse. The broad absorption band observed at 3335 cm^{-1} corresponds to O–H stretching vibrations, indicating the presence of hydroxyl groups typical of cellulose. The peak at approximately 2940 cm^{-1} is attributed to C–H stretching of aliphatic groups. The characteristic absorption band near 1030 cm^{-1} corresponds to C–O–C stretching vibrations of the *β*

Conclusion

The solution casting method successfully fabricated curcumin-loaded HEC/PVA/SA biocomposite hydrogels using sugarcane bagasse cellulose as a sustainable raw material. FTIR and FESEM analyses confirmed the effective incorporation of curcumin particles within the biocomposite matrix. Although the HEC/PVA hydrogels exhibited the highest swelling capacity, this property decreased with increasing curcumin content. The synthesized biocomposites demonstrated strong antibacterial activity against *S.*

CRediT authorship contribution statement

Md Hasibul Islam: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. **Md Ibrahim H. Mondal:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.